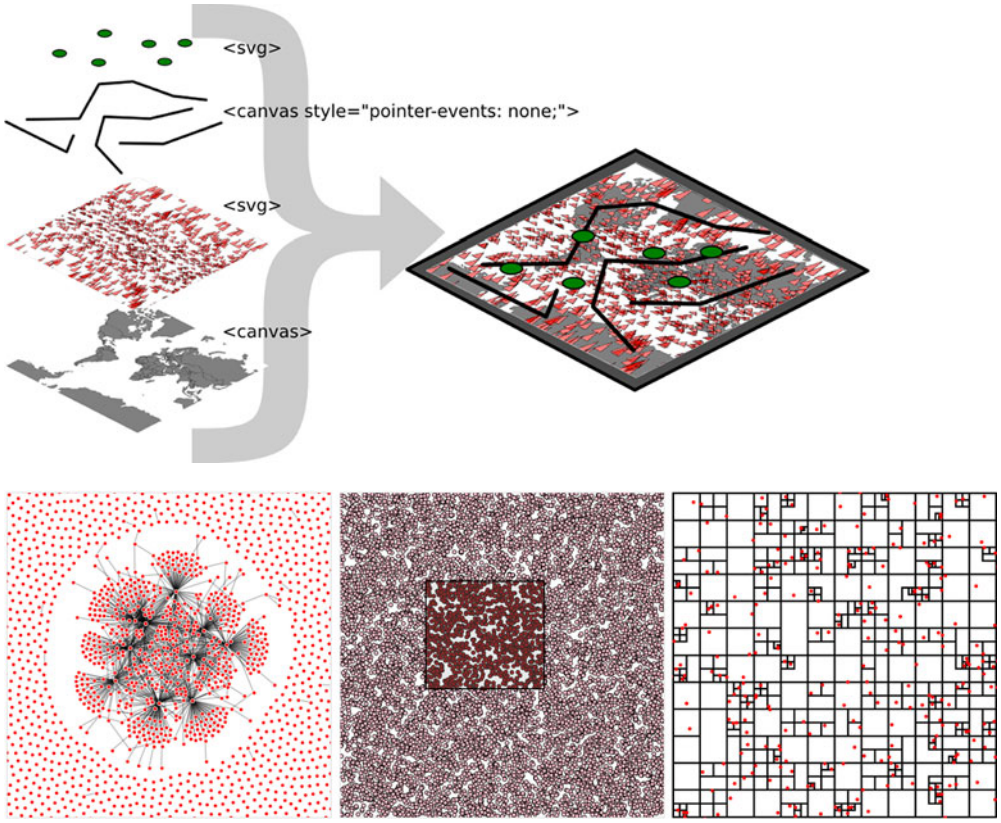


Figure 11.1 This chapter focuses on optimization techniques such as using HTML5 canvas to draw large datasets in tandem with SVG for the interactive elements. This is demonstrated with maps (section 11.1), networks (11.2), and traditional xy data (section 11.3), which uses the D3 quadtree function (section 11.3.2).

and performance. We'll also revisit HTML5 canvas throughout the chapter to see how we can use canvas in tandem with SVG to get the high performance and maintain the interactivity that SVG is so useful for.

We've worked with data throughout this book, but this time, we'll appreciably up the ante by trying to represent a thousand or more datapoints using maps, networks, and charts, which are significantly more resource-intensive than a circle pack chart, a bar chart, or a spreadsheet.

11.1 Big geodata

In chapter 7, you had only 10 cities representing the entire globe. That's not typical: when you're working with geodata, you'll often work with large datasets describing many complex shapes. Fortunately, there's built-in functionality in D3 for drawing that complex data with HTML5 canvas, which dramatically improves performance. For this chapter, we'll need to include a `<canvas>` element in our DOM.